

GAR, Russian Federation

WMO: 313840

Lat: 52.567N

Lon: 129.067E

Elev: 807

StdP: 14.27

Time Zone: 9.00 (E09)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.5	-41.1	-47.8	0.3	-44.3	-43.9	0.4	-40.7	15.3	-1.2	11.1	-2.5	0.0	N/A	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	83.7	68.4	81.4	68.1	78.3	65.8	72.4	78.7	70.2	77.1	68.4	75.4	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.5	115.5	75.7	68.2	106.6	73.4	66.1	99.1	71.3	36.7	79.1	34.9	77.0	33.3	75.4	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.9	7.8	-49.5	90.4	4.9	3.3	-53.0	92.8	-55.9	94.8	-58.6	96.6	-62.1	99.1
			-48.5	77.7	3.7	4.2	-51.2	80.7	-53.4	83.1	-55.4	85.5	-58.1	88.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	-19.5	-6.3	11.3	32.7	47.4	61.4	65.2	60.8	47.4	31.2	1.7	-16.0
	DBStd	30.99	9.01	9.05	11.07	8.53	7.98	5.43	4.91	5.58	7.01	8.68	10.52	9.45
	HDD50	9813	2155	1576	1199	520	147	2	0	1	134	583	1449	2047
	HDD65	14131	2620	1996	1664	969	545	134	59	155	530	1048	1899	2512
	CDD50	1271	0	0	0	0	67	343	472	335	54	0	0	0
	CDD65	116	0	0	0	0	0	25	66	24	1	0	0	0
	CDH74	1816	0	0	0	0	87	549	822	351	7	0	0	0
	CDH80	367	0	0	0	0	15	116	172	63	1	0	0	0
Wind	WSAvg	2.0	0.9	1.3	2.2	3.5	3.5	2.4	2.0	2.0	2.0	2.1	1.6	0.7
Precipitation	PrecAvg	22.2	0.3	0.2	0.4	0.9	2.4	3.1	5.2	4.7	2.6	1.2	0.7	0.4
	PrecMax	31.4	0.9	0.6	1.8	2.3	5.3	5.4	10.7	8.0	4.8	2.9	1.8	0.9
	PrecMin	16.1	0.0	0.0	0.0	0.1	0.6	1.4	1.7	1.3	1.1	0.2	0.1	0.1
	PrecStd	4.2	0.2	0.2	0.4	0.6	1.3	1.1	2.6	1.8	1.0	0.6	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	25.6	50.9	62.8	82.2	87.0	86.7	85.3	74.6	60.3	32.7	14.3
		MCWB	10.5	22.1	41.2	47.5	62.3	66.9	70.6	73.6	62.1	46.5	28.9	13.4
	2%	DB	6.3	20.8	40.6	58.4	75.3	83.1	83.9	80.9	70.7	55.5	27.7	9.6
		MCWB	5.3	17.8	33.0	44.3	57.3	66.0	70.1	70.1	59.2	44.4	24.8	8.8
	5%	DB	1.1	15.7	35.4	53.2	70.5	80.3	81.8	77.7	66.9	51.4	24.8	5.0
		MCWB	0.1	13.0	29.4	40.7	54.3	65.2	69.6	66.4	58.0	41.5	22.5	4.1
	10%	DB	-2.8	11.3	30.8	49.1	65.5	77.0	78.9	74.6	62.7	47.2	20.6	0.7
		MCWB	-3.6	9.1	25.9	38.1	50.9	63.7	68.2	65.1	54.6	40.0	18.5	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	22.9	41.2	48.6	62.2	71.3	78.1	77.3	65.4	48.8	29.5	13.6
		MCDB	12.3	25.4	51.2	60.9	77.2	80.7	82.3	82.9	70.7	57.3	32.4	14.4
	2%	WB	5.5	18.0	34.1	44.4	58.3	69.6	73.6	71.1	62.0	45.6	25.4	8.9
		MCDB	6.5	20.8	39.2	56.5	71.2	78.8	79.5	77.5	67.8	52.5	27.5	9.7
	5%	WB	0.2	13.1	29.7	41.6	55.8	67.9	71.4	68.6	59.1	42.5	22.2	4.2
		MCDB	1.0	15.7	35.0	51.9	68.8	75.8	77.5	74.6	63.9	50.5	24.6	5.0
	10%	WB	-3.5	9.3	26.3	38.8	52.6	66.0	69.6	66.4	56.0	40.0	18.4	0.0
		MCDB	-2.8	11.4	30.3	47.8	63.3	73.2	76.2	72.8	61.9	46.6	20.6	0.7
Mean Daily Temperature Range	5% DB	MDBR	28.9	33.9	31.4	24.6	26.8	26.3	23.5	23.8	25.8	25.5	25.5	25.9
		MCDBR	27.9	36.2	33.0	30.8	36.8	34.8	29.4	30.0	31.3	33.1	24.9	26.5
		MCWBR	26.8	32.9	25.5	20.2	21.3	18.5	17.3	18.5	21.0	22.4	22.0	25.4
	5% WB	MCDBR	28.0	36.0	31.4	28.7	33.7	25.6	24.1	25.8	27.2	30.5	23.2	26.4
		MCWBR	27.0	32.8	24.5	19.8	19.8	16.0	17.5	18.1	20.4	21.8	21.2	25.2
Clear-Sky Solar Irradiance	taub	0.255	0.289	0.355	0.536	0.587	0.500	0.474	0.418	0.388	0.395	0.311	0.256	
	taud	2.146	2.084	1.952	1.691	1.686	1.998	2.125	2.276	2.273	2.056	2.093	2.096	
	Ebn at Noon	232	256	257	220	217	242	246	255	248	208	198	203	
	Edh at Noon	22	32	45	68	72	53	46	37	33	33	23	19	
All-Sky Solar Radiation	RadAvg	441	819	1362	1631	1689	1922	1719	1493	1131	738	466	339	
	RadStd	16	23	58	161	124	183	157	100	90	53	24	15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page